

PROJECT SYNOPSIS

Drone Based Human Drowning Detection Using Image Processing

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Introduction

Drowning is the third leading cause of unintentional injury & death worldwide. 1.2 million people around the world die and get injured by drowning every year, more than 2 people per minute.

Supervised beaches are the most challenging environments to lifeguards, due to the influence of external factors such as weather, currents, tides, waves, and poor visibility. On the other hand, unsupervised beaches are death traps since there is no one to educate and manage the crowd or to assist and alert in case of an emergency.

Solution to our problem:

Drone or unmanned aircraft systems (UAS) technology has become a powerful tool in public safety.

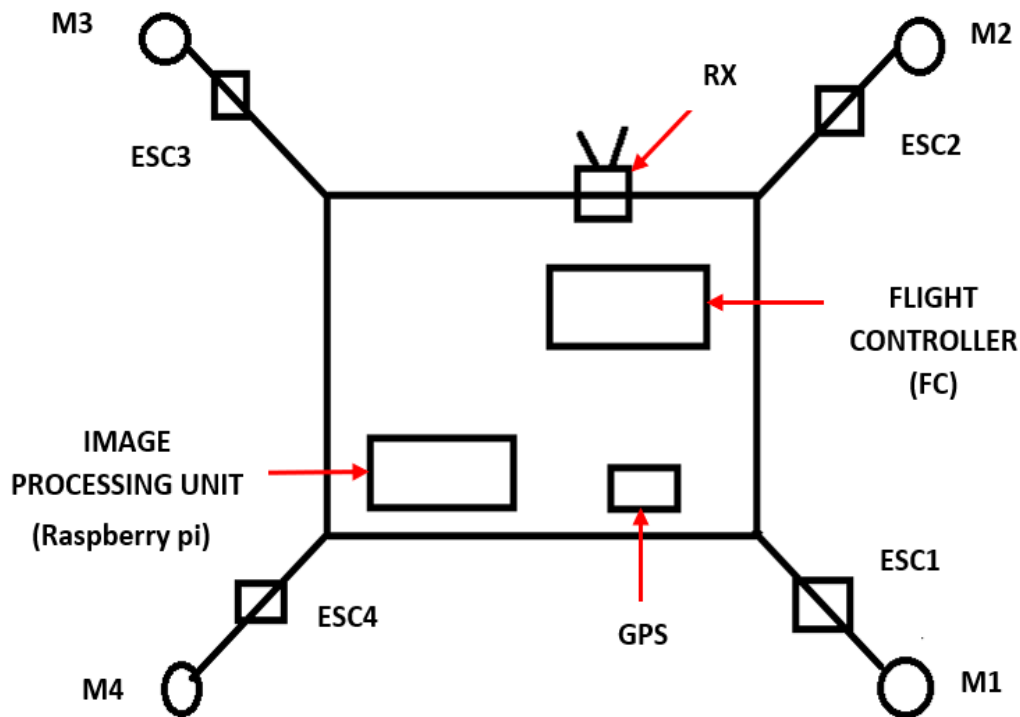
By using a combination of a drone and various image processing techniques we are trying to overcome this problem. This drone along with the image processing unit will help to locate people drowning in water and get an approximation of the location they are trapped in and notify the concerned authority about the same. Also, it will keep an eye on the people breaking the rules and regulations which are supposed to be followed strictly for their own safety.

Block Diagram

This block diagram is a brief structural overview of the project. There is a Quadcopter with four brushless motors (denoted by symbols, M). These motors are controlled by the electronic speed controllers (ESCs). ESCs are responsible for regulating the speed of each individual motor.

The body of the drone consists of a flight controller, GPS and RX module, battery and image processing unit. The flight controller is the central brain of the drone. It is responsible to control everything the drone does. The board which will be used is the APM board. The GPS module will send the position of the drone to the controller. While the Rx module will receive the inputs from the controller regarding the motion or direction in which the drone will move. These inputs are

given to the flight controller which will process these inputs and execute by giving



the commands to the ESCs. Battery supply is needed to supply power to all the components of the drone. Addition of a power distribution board will add benefit so as to distribute the power evenly in all the four motors and other components.

The image processing unit which is our highlight of the project will consist of a Raspberry PI board and PI camera which will capture images and GPS coordinates which will be sent to the user at the controller.

Specification

Quadcopter Hardware Parts	Cost
1 x Quadcopter Frame + propellers	700
4 x Brushless Motor	1600
4 x ESC	2000

1 x Flight Controller (APM)	2000
1 x Raspberry Pi	11-13k
1 x Pi Camera	350
1 x GPS	800
1 x Flysky controller FS-16	4000
2 x LiPo	2000
1x IMAX B3 AC Compact balance CHARGER for LIPO	500

Approximate Budget: 20,000/- to 25,000/-

Technologies

- ❖ Python : OpenCV
 - Computer vision & image processing
 - Detecting edges, sharpening, blurring, cropping, contouring, object detection.
- ❖ Python: Raspbian OS (Raspberry Pi) (Debian version)
 - Numpy
 - Matplotlib libraries.
- ❖ C/Python language for serial communication between Raspberry Pi & flight controller
 - serial console and serial Port
 - Rx & Tx
- ❖YOLO

Features:

- GPS tracking of the drone.
- Image processing.
- Autopilot search and rescue.
- Alarm / Alerting system.

- The camera's location captures a complete view of the facility, including swimmers, wanderers, and occupied objects.
- Auto Return home when battery is about to discharge

Optional Features That can be Added (if time permits):

- Night Vision camera for surveillance purposes
- LED lighting so that the robot acts as a beacon in the dark.
- Investigating other emergency situations such as floods and earthquakes.
- Detecting large, dangerous sea animals in proximity.

Summary of the project

In this project, the drone will be able to detect the person drowning in the sea, following which images and the location of the person will be captured with the help of an image processing unit. This captured image along with the coordinates of the drone will be displayed on the screen. Hence the user at the controller will be able to detect the person.

Also the image processing unit will also send a notification to the specified authority. This drone can decrease the duration of rescue operations and save more lives.